Deep Learning: A Retrospective

Reading LeCun, Bengio, and Hinton (Nature, 2015)

Your Name

STAT 3494W

April 24, 2026

Outline

1. Introduction
2. Why Does This Paper Exist?
3. The Basics: Deep Learning
4. CNNs and ImageNet
5. RNNs and Language
6. What They Predicted
7. Then Transformers Happened
8. Conclusion
9. References

The Paper

- **Published:** Nature, 2015
- **Authors:** LeCun, Bengio, Hinton
- **Citations:** 70,000+
- Most prestigious scientific journal in the world

"Today we're going to read this paper the way it was meant to be read — as an argument, not just a summary."

Why Should We Trust These People?

- **Three researchers, three legends**
- **Turing Award 2018** — for their work on deep learning
- **Nobel Prize in Physics 2024** — Hinton & Hopfield

"We'll meet each of them properly as we go..."

The AI Hype Cycle

- **1950s–60s:** Hype cycle 1 — early AI optimism
- **1970s:** AI Winter 1 — symbolic AI hits limits
- **1980s:** Expert systems — hype cycle 2
- **1990s:** AI Winter 2 — expert systems fail
- **2000s:** ML revival — SVMs, kernel methods dominate
- **2012:** Something changes...

The Skeptic's View

By 2015, deep learning is working — but the research community remembers the winters.

> *"Thought learning useful, multistage, feature extractors with little prior knowledge was infeasible."*

This is the audience the paper is written for.

What the Paper Is Actually Doing

- **Not** just a technical summary
- A **persuasion document** written by true believers with receipts
- They're selling something they *know* is real

"Let's see if they were right..."

Road Map

1. **The Basics** — Backpropagation
2. **CNNs** — Convolutional Neural Networks
3. **RNNs** — Recurrent Neural Networks
4. **Predictions** — What did they predict?
5. **Transformers** — What happened next?

The Core Idea

- Replace **hand-engineered features** with **trainable multilayer networks**
- Why this matters: feature engineering was the bottleneck

```
Raw Input → Layer 1 → Layer 2 → ...  → Output
```

Meet Geoffrey Hinton

- Co-author of the **1986 backpropagation paper**
- Invented **Boltzmann machines** (1985)
- **Nobel Prize in Physics 2024**

"One of the authors literally helped invent what we're about to explain."

Backpropagation

1. **Forward pass:** Input $\rightarrow$ through layers $\rightarrow$ output
2. **Compute loss:** How wrong are we?
3. **Backward pass:** Propagate gradients back through layers
4. **Update weights:** Adjust in direction that reduces error

ReLU: Fastest activation function — what the paper recommends

Stochastic Gradient Descent

- The optimization engine
- Discovered by multiple groups in the 70s and 80s
- Including Hinton's group

"This simple procedure usually finds a good set of weights surprisingly quickly."

"But Does it Actually Work?"

The saddle point rebuttal:

> *"The landscape is packed with a combinatorially large number of saddle points where the gradient is zero, and the surface curves up in most dimensions and curves down in the remainder."*

Direct answer to the skeptics.
The authors know exactly what objection you just had.

The 2006 Breakthrough

- **CIFAR group** — first model to detect features *without labeled data*
- Worked around limitations: small datasets, no ReLU, poor GPU support
- **Proved deep learning was worth pursuing**

Reveal: The authors don't even mention that *they were the ones who did this.*

Led to first commercial win: **speech recognition on Android by 2012**

Meet Yann LeCun

- **Invented CNNs** (LeNet)
- Bank check recognition system
- Reading **10% of all US checks** by late 1990s

"Another author. Another invention we're about to explain."

What is a CNN?

The problem: Standard networks ignore spatial structure

The solution: Convolutional Neural Networks

- Convolution layers — detect local features
- Pooling layers — reduce spatial dimensions
- Feature maps — hierarchical representations

Biologically inspired — mirrors the visual cortex

How CNNs are Trained

"When ConvNet models and monkeys are shown the same picture, the activations of high-level units in the ConvNet explains half of the variance of random sets of 160 neurons in the monkey's inferotemporal cortex."

They're not just saying it works — they're saying it works like a brain.

Early CNN Deployments

- Speech recognition, OCR, handwriting recognition
- Microsoft deployments
- Face and hand detection in the early 90s

"This was working for 20 years and the mainstream ignored it."

ImageNet 2012

AlexNet enters the competition

- **Almost halving** error rates of the best competing approach
- Key ingredients:
  - GPUs
  - ReLU activation
  - Dropout regularization
  - Data augmentation

This is the moment the mainstream could no longer ignore it.

The World Notices

"Companies such as Mobileye and NVIDIA are using such ConvNet-based methods in their upcoming vision systems for cars."

- NVIDIA investing in deep learning chips
- Tesla self-driving cars
- *If you read this paper in 2015 and bought NVIDIA stock...*

The hype is warranted — they know it's going to work.

Why Was it Ignored for So Long?

- CNNs were largely forsaken until 2012
- **Despite working since the early 90s**
- The AI winters left scars

"If it works for images, what about language?"

Meet Yoshua Bengio

- **Most cited computer scientist globally**
- Most cited living scientist across all fields
- Neural probabilistic language model — **word embeddings**

"Third author. You know what's coming."

The Language Problem

Old approach: N-grams — counting symbol frequencies

Number of possible N-grams: V^N where V = vocabulary size

- Brittle
- Slow
- No understanding of meaning

The Exponential Advantages of Deep Networks

Two reasons deep networks are powerful for language:

1. n binary features $\rightarrow 2^n$ possible combinations
   "30 neurons can represent over a billion concepts"
2. Each layer **multiplies** expressive power exponentially

Word Embeddings

- **One-hot encoding:** one component $= 1$, rest $= 0$
- First layer creates activation patterns per word
- Foreshadows Word2Vec and modern embeddings

Example:

- Tuesday and Wednesday $\rightarrow$ similar vectors
- Sweden and Norway $\rightarrow$ similar vectors

"They didn't program it to learn this. It just did."

Simple Feed Forward Language Model

- Given 5 words, predict the next
- Works surprisingly well on:
  - Small, clean datasets
  - Large, messy real-world corpora

This was contrary to what people expected.

Recurrent Neural Networks (RNNs)

The problem: Feed-forward networks can't handle sequences

- Can translate English to French
- Learns semantic rules of both languages
- **Problem:** Vanishing gradient

Long Short-Term Memory (LSTM)

Invented 1997

- Cell state — long-term memory
- Gates — control information flow
- **Intuition:** Memory with a forget switch

The authors are bullish — and they're right.

The Memory Bottleneck

The biggest remaining problem:

- Neural networks are **stateless**
- Memory is inherently **stateful**

Neural Turing Machines (DeepMind, 2014)

- Differentiable Turing machine
- Attention-based read/write to memory

"They've identified the exact problem. The solution is two years away."

The Paper's Bets

2015. Let's grade their predictions.

1. Unsupervised learning
2. Attention mechanisms
3. Reinforcement learning for vision
4. Natural language understanding

Prediction: Unsupervised Learning

They said: Training largely unsupervised is the path forward.

What happened:

- GPT — unsupervised pretraining
- BERT — self-supervised learning
- Modern LLMs

Grade: ✓

Prediction: Attention

"Systems that use RNNs... will become much better when they learn strategies for selectively attending to one part at a time."

What happened:

- "Attention is All You Need" — 2017
- Transformers

Grade: ✓✓

Prediction: Reinforcement Learning for Vision

"Human vision is an active process that sequentially samples the optic array in an intelligent, task-specific way."

What happened:

- CNN + RNN + RL
- DeepMind's game-playing agents

Grade: ✓

Prediction: Natural Language Understanding

"Major progress will come through systems that combine representation learning with complex reasoning."

What happened:

- LLMs are impressive
- But still brittle on reasoning

Grade: Partial ✓— making progress, not solved

2017: A New Architecture

Transformers

- You've seen this before in class
- "Attention is All You Need" discards recurrence entirely

"Now you know what problem it was solving."

What Transformers Got Right That the Paper Missed

- The paper assumed RNNs were the right backbone for sequences
- Transformers showed: **you don't need recurrence at all**
- If attention is powerful enough

The memory bottleneck they identified was real — but the solution wasn't Neural Turing Machines, it was **self-attention**.

What the Paper Got Right About Transformers

(Without knowing it)

- **Unsupervised pretraining:** GPT is exactly this
- **Attention as key mechanism:** they named it
- **Scale matters:** they said it, transformers proved it

"They predicted the destination without predicting the vehicle."

Conclusions That Still Hold

- Deep learning over hand-engineered features: **obviously true**
- Scale + data + compute: **validated completely**
- Representation learning as foundation: **yes**

Still open: Reasoning, grounding, true language understanding

Where Are We Now

- LLMs are extraordinary — but not AGI
- The paper's philosophical question still stands:

  "New paradigms are needed to replace rule-based manipulation of symbolic expressions."

We have better tools. The question isn't answered.

What Made This Paper Special

- Not just a summary — a **persuasion document**
- Written by the people who **built the field**
- Every claim backed by work they personally did
- They were selling something they knew was real

The Scorecard

Prediction	**Outcome**
Unsupervised learning	✓
Attention mechanisms	✓✓
RL for vision	✓
Language understanding	Partial ✓
RNNs as backbone	× (Transformers won)

Thank You / Questions

"They predicted the future pretty well.
The parts they got wrong are the parts we're still working on."

References